column-hse-ltau: hydrostatic columns on an optical-depth grid

The physics of `run.mode = "column-hse-ltau"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

When we observe the Sun we do not see heights; we see light. A Stokes inversion therefore returns the temperature and the magnetic field not at fixed geometric heights but on surfaces of constant **optical depth** τ_{500}: at $\log_{10}\tau_{500} = 0, -1, -2, \ldots$ in every pixel. To turn such an atmosphere into a physical one we need, in every pixel, the gas pressure and the density on each τ surface, and the **geometric height** at which that surface lies. This mode computes them by assuming that every pixel column is in hydrostatic equilibrium on its own.

2. The quantities

Symbol	Meaning	Unit
τ	optical depth at 500 nm, counted from the top downward; the grid levels are equally spaced in $\log_{10}\tau$	–
T, $\mathbf{B}$	temperature and magnetic field on the τ levels, given, never changed	K, T
P, ρ	gas pressure and density on the τ levels	Pa, $\mathrm{kg\,m^{-3}}$
χ	extinction coefficient at 500 nm (fraction of the light removed per metre)	$\mathrm{m^{-1}}$
κ	extinction per unit mass, $\kappa = \chi/\rho$	$\mathrm{m^2\,kg^{-1}}$
m	column mass: mass of gas above a point per unit area	$\mathrm{kg\,m^{-2}}$
$z(\tau)$	geometric height of a τ level in a column	m
g	solar surface gravity, 274	$\mathrm{m\,s^{-2}}$
μ_0	vacuum permeability, $4\pi \times 10^{-7}$	$\mathrm{T\,m\,A^{-1}}$

3. Optical depth is a measure of mass

Optical depth is defined by how much light is removed on the way out: going down a distance dz adds $d\tau = \chi\,dz$ (with z upward, $d\tau = -\chi\,dz$). Write $\chi = \kappa\rho$: the extinction is the product of how much gas there is and how strongly each kilogram of it absorbs. Then

$$d\tau = -\kappa\,\rho\,dz = \kappa\,dm,$$

because $-\rho\,dz = dm$ is the mass added to the column above as we go down by dz. So optical depth is *column mass weighted by the opacity*. For a gas with constant κ, $\tau = \kappa\,m$ exactly.

4. Hydrostatic equilibrium in optical depth

On a height grid, hydrostatic equilibrium reads $dP/dz = -\rho g$ (the weight of a thin slab is carried by the pressure difference across it). Divide it by $d\tau = -\kappa\rho\,dz$:

$$\boxed{\frac{dP}{d\tau} = \frac{g}{\kappa}}$$

This is the classical form used in stellar atmospheres. Since $dP = g\,dm$ (the pressure is the weight of the gas above), it simply says that adding the optical depth $d\tau$ means adding the mass $d\tau/\kappa$. With constant κ it integrates to $P = g\tau/\kappa$: *the pressure at optical depth τ is the weight of the gas that produces that much opacity.*

The levels are equally spaced in $\log_{10}\tau$, so the code integrates in $\ln\tau$ ($d\tau = \tau\,d\ln\tau$) and in $\ln P$ ($dP = P\,d\ln P$):

$$\boxed{\frac{d\ln P}{d\ln\tau} = \frac{g\,\tau}{\kappa\,P} = \frac{g\,\tau\,\rho}{\chi\,P}}$$

ρ and χ depend on T and P through the equation of state (ideal gas with ionisation and molecules, from Wittmann's code, tabulated in $(\ln T, \ln P)$). The equation is therefore integrated numerically, level by level from the top down.

5. Why no reference pressure is needed

On a height grid the pressure needs a prescribed value somewhere (an anchor), because a height coordinate says nothing about how much gas there is: the same temperature profile is equally hydrostatic at any pressure level. Optical depth is different: τ measures the mass above a point, and hydrostatic equilibrium says that the pressure *is* the weight of that mass. Once $T(\tau)$ and the opacity of the gas are known, the pressure is fixed; there is nothing left to choose. The flip side is that the result is only as good as the opacity: if the τ scale of the input was computed with a different opacity, the pressures differ in proportion.

6. The top of the column

The integration needs a starting value at the top level τ_1 (here 10^{-8}). Above it we assume κ stays constant at its value κ_1 and that the pressure vanishes at $\tau = 0$. Then the equation integrates to

$$P_1 = \frac{g\,\tau_1}{\kappa_1(T_1, P_1)}.$$

Because κ_1 itself depends on P_1, this is solved by iteration (guess P_1, compute κ_1, update P_1, repeat). An error in this assumption matters only near the top: deeper down the pressure is dominated by the gas added below τ_1. Roughly, its relative weight is τ_1/τ, i.e. 10^{-8} at $\tau = 1$.

7. Heights from the opacity

A τ level has a height only once we know how thick each layer is. From $d\tau = -\chi\,dz$:

$$\boxed{\frac{dz}{d\ln\tau} = -\frac{\tau}{\chi}}$$

Where the gas is opaque (large χ), a step in $\ln\tau$ is a thin layer; where it is transparent, the same step is a thick layer. z is integrated alongside $\ln P$. This fixes the heights in each column only *up to an additive constant*: we know the distance between the levels, not where the column sits (Sect. 9).

8. Between grid levels: hot gas over cool gas

The grid has 10 levels per factor of ten in τ. Between two levels $\ln T$ is assumed to vary linearly with $\ln\tau$, and the integration takes small steps so that $\ln P$ changes by at most 0.05 per step.

The opacity at 500 nm comes mainly from the negative hydrogen ion H^-, which needs free electrons, and both rise very steeply with temperature. So cool gas is far more transparent than hot gas. Suppose a hot layer (8000 K) lies directly on a cool one (4000 K) within one grid interval. With the linear interpolation, a whole tenth of a decade in τ is then filled with nearly transparent gas. By $dP/d\tau = g/\kappa$, collecting that much optical depth from transparent gas requires a huge amount of mass, and the pressure jumps by orders of magnitude within the interval. This is a consequence of representing a sharp temperature change by smooth interpolation on a coarse grid, not of the physics. The code therefore limits the rise of $\ln P$ to 5 per 0.1 dex in τ (`cap_per_dex` = 50 per dex). This limit is far above the rises of normal stratifications, so they are untouched, and below a limited interval the pressure returns to the hydrostatic value within a few levels.

9. Where the columns sit: total-pressure alignment

9.1 Magnetic pressure

In each column we now know P, ρ and the layer thicknesses, but not the height of the column as a whole. The mode fixes it with a physical argument that involves the magnetic field. The force a magnetic field exerts on the gas, the Lorentz force, can be rewritten with a vector identity as

$$\mathbf{J}\times\mathbf{B} = \frac{(\nabla\times\mathbf{B})\times\mathbf{B}}{\mu_0} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0}.$$

The first term acts like an additional pressure, the **magnetic pressure** $B^2/2\mu_0$; the second is a tension along curved field lines.

9.2 The alignment rule

Deep in the atmosphere neighbouring columns push on each other until the *total* pressure, $P + B^2/2\mu_0$, is the same at the same height (if it were not, the higher total pressure would push the neighbour aside). The mode uses this as its alignment rule: it takes as reference the median, over all columns, of the total pressure on the deepest τ level, and shifts every column vertically so that its total pressure reaches the reference at one and the same height. A column whose deepest level has a total pressure below the reference is continued downward with the isothermal hydrostatic law $P \propto e^{-z/H}$ until it reaches it.

9.3 Consequence: the Wilson depression

A column with a strong field carries part of the load by magnetic pressure, so it needs less gas pressure at the reference height. Less gas means a lower density, a lower opacity and a longer path to reach $\tau = 1$: its $\tau = 1$ surface lies deeper. Sunspots seen near the limb show exactly this, the Wilson depression.

9.4 The gauge

Finally all columns are shifted together so that the average height of $\tau = 1$ is zero. This is only a choice of origin, with no physical content.

10. What the result means

- Every column is in hydrostatic equilibrium along τ: the pressure on each τ level is the weight of the gas above it.
- The heights follow from the opacity; the field enters only through the alignment, via its magnetic pressure.
- The columns are not in horizontal force balance with each other, and the tension of the field is ignored. Adding both is the purpose of `mhs-poisson-ltau`, which starts from this result.